

Algebra 1
Unit 2
Lesson 7 Homework

Name Answer Key
Date _____
Period _____

1. Find the product of $(x^2 - 3)(4x + 2)$ using the distributive property.

$$4x^3 + 2x^2 - 12x - 6$$

2. Find the product of $(x^2 - 3)(4x + 2)$ using an area model.

	x^2	-3
$4x$	$4x^3$	$-12x$
$+2$	$2x^2$	-6

$$4x^3 + 2x^2 - 12x - 6$$

3. Find the product of $(-\frac{1}{4}y + 3y^2 - 7)(2y - 5)$ using the distributive property.

$$-\frac{1}{2}y^2 + \frac{5}{4}y + 6y^3 - 15y^2 - 14y + 35$$

$$6y^3 - 15\frac{1}{2}y^2 - 12\frac{3}{4}y + 35$$

4. Find the product of $(-\frac{1}{4}y + 3y^2 - 7)(2y - 5)$ using an area model.

	$-\frac{1}{4}y$	$3y^2$	-7
$2y$	$-\frac{1}{2}y^2$	$6y^2$	$-14y$
-5	$\frac{5}{4}y$	$-15y^2$	35

$$6y^3 - 15\frac{1}{2}y^2 - 12\frac{3}{4}y + 35$$

Find each product.

5. $(-6a + \frac{2}{5}b)^2$

$$36a^2 - \frac{24}{5}ab + \frac{4}{25}b^2$$

6. $(2.4 - 3c)(-0.3 + 5c^3 - c^2)$

$$-15c^4 + 15c^3 - 2.4c^2 + 0.9c - 0.72$$

7. $(\frac{3}{4}x - 7)(\frac{-2}{5} + 8x)$

$$6x^2 - 56.3x + 2.8$$

8. $(-2\frac{1}{2} + 5d^2)(6d - 6.1c - 3d^2)$

$$-15d^4 + 30d^3 + 7.5d^2 - 30cd^2 + 15.25c - 15d$$

9. $(12 + 6m)(-8n - \frac{18}{5})$

$$-48mn - 21.6m - 96n - 43.2$$

10. $(4r^2 - \frac{7}{4} + 19r)(-3r + 2r^2 + 15)$

$$8r^4 + 26r^3 - 0.5r^2 + 290.25r - 26.25$$